

- **Positive change in metric**
- **p value < 0.05**
- **Can we stop the experiments?**

DATA PEEKING

- Stop collecting data when the test comes out significant
- Result is **inaccurate**

DATA PEEKING

When calculate duration of an experiment

- statistical power
- significance level
- day of week effect
- seasonality
- etc

DATA PEEKING

Stop experience earlier than designed

- Estimated treatment effect $\neq$ true treatment effect

DATA PEEKING

Goal of A/B tests

- result is **reproducible**

Stop tests prematurely

- result won't be reproducible

DATA PEEKING

What should we do?

- Run the experiment as long as designed

When does it occur?

- **multiple metrics** in an A/B test
- one metric in an a/b test with **multiple treatment groups**
- a **segment** of the population

Eg. use platform to slice the population

- Users groups: **web**, **Android**, **iOS**, etc.
- This is a multiple testing problem

When does it occur?

- **multiple metrics** in an A/B test
- one metric in an a/b test with **multiple treatment groups**
- a **segment** of the population
- **multiple iterations** of an A/B test
- **multiple a/b tests** in parallel

Two-step rule of thumb

- separate all metrics into **three** groups
 - those you expect to be impacted
 - those potentially to be impacted
 - those unlikely to be impacted
- apply **tiered significance levels** to each group

Source: trustworthy online controlled experiments (P119)

Experiment metrics: A, B, C, D

- Metric A: expected to be impacted
- Metric B, C: not sure
- Metric D: expected NOT to be impacted

Experiment metrics: A, B, C, D

- Metric A → group 1, sig. Lvl = 0.05
- Metric B, C → group 2, sig. Lvl = 0.01
- Metric D → group 3, sig. Lvl = 0.001

Experiment metrics: A, B, C, D

- Metric A → Significant
- Metric B, C → NOT Significant
- Metric D → NOT Significant

GREAT NEWS!

Experiment metrics: A, B, C, D

- Metric A → NOT Significant
- Metric B, C → Significant
- Metric D → Significant

Experiment metrics: A, B, C, D

- Metric A → NOT Significant
- Metric B, C → Significant
- Metric D → Significant

Look into source of error

- **Change in conversion rate is not statistically significant**
- **Claim not treatment effect**

- Test may be **underpowered** to detect the effect size
 - Not enough randomization units in the test
- Expect 1000 unique users but have 900
 - Experiment design phase
 - Duration based on the **estimated** number of randomization units
 - Estimated number of user may be larger than real number

Less users than desired

- Customers churned
- Expect 1000 unique users but have 900
- Draw conclusion: NO treatment effect

- What to do?
 - Make sure experiment has enough statistical power
 - Continue running the experiment
 - Rerun the experiment